Exploring AI with Google AI Studio

Today's goal • 3/3 items

Module 1
Introduction to AI Studio

- Explore AI Studio
  Video • 1 min
- Your gateway to Google's AI ecosystem
  Video • 2 min
- AI Studio overview and capabilities
  Reading • 10 min
- Hands-on feature exploration
  Video • 3 min
- Activity: Feature discovery
  Practice Assignment • Grade: 100%
- How you could use AI Studio in your role
  Dialogue • 15 min
- Knowledge check: AI Studio overview
  Practice Assignment • Grade: 100%

Module 2
Prompt Engineering and Model Evaluation

- Working with your model to get the best results
  Video • 2 min
- Prompting, models, and customization
  Video • 4 min
- **Prompt and model evaluations**
  Reading • 10 min
- Testing and refining output in AI Studio
  Video • 4 min
- Activity: Optimizing your AI requests
  Practice Assignment • 30 min
- Optimization strategies
  Dialogue • 15 min
- Knowledge check: AI customization and optimization
  Practice Assignment • 15 min

Module 3
Vibe Coding with AI Studio Build Mode

- Go from a single prompt to an AI-powered app
  Video • 1 min
- Creating and exporting code for your application
  Reading • 10 min
- Coding with AI
  Video • 3 min
- Activity: Converting your prototype into an application
  Practice Assignment • 30 min
- Application development workflow
  Dialogue • 15 min
- Graded quiz: Building your application in AI Studio
  Graded Assignment • 30 min
- Course wrap-up
  Reading • 10 min

Prompt and model evaluations

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Understanding the need for structured evaluation

Getting a good response from an artificial intelligence model can sometimes seem unpredictable. You might receive an excellent answer one moment and an unhelpful one the next, even with similar inputs. This inconsistency is a key challenge in building reliable AI applications.

Instead of relying on repeated trials and luck, professional AI development uses a structured, scientific approach to testing and refining how we communicate with models. You can transform unpredictable outcomes into consistent, high-quality results by defining what a successful output looks like, setting a clear starting point for comparison, and using systematic methods to test changes. This process makes prompt engineering a repeatable skill—essential for building effective, scalable AI applications.

Defining success criteria

The first step in any evaluation is to define success. Without clear goals, it's impossible to measure whether a prompt or model performs well. Success criteria are the specific standards used to assess the quality of an AI's output, and they vary by use case.

For example, in a summarization task, success might depend on accuracy—whether the summary correctly reflects the main facts of the source text. In contrast, for a communication task, the key measure could be tone—whether the output sounds natural and aligns with the intended style.

Other commonly used criteria include:

- **Relevance:** Is the response directly related to the input query?
- **Safety:** Does the output avoid harmful or biased content?
- **Efficiency:** How quickly and cost-effectively was the response produced?

By setting these measurable criteria upfront, you gain a clear standard against which every AI response can be evaluated. Defining these metrics early also helps ensure alignment between technical goals and user expectations, creating a consistent foundation for future testing and refinement.

Applying success criteria through an example

Consider a scenario where a developer needs to generate short, clear content using an AI model. Before writing any prompts, the team first defines what success looks like. They decide on three clear criteria:

1. The response must stay within a defined word limit.
2. It must include all required information or keywords.
3. The tone must be appropriate for the intended audience.

With these measurable goals in place, the developer now has a clear framework to evaluate every AI-generated output. Any response that exceeds the limit, misses key information, or uses an inappropriate tone can be identified as ineffective—regardless of how well written it appears.

Establishing a baseline prompt

Once success is defined, the next step is to create a baseline prompt—the simplest, most direct version of the task. This prompt provides a starting point for comparison and helps track improvement as refinements are made.

For example, a developer might start with a straightforward instruction such as:

"Summarize the main points of this document."

The model's output could be:

"The document explains the importance of maintaining data security."

While this response is factually correct, it may be too brief and not fully aligned with the success criteria—for instance, lacking clarity, tone, or completeness. Still, it serves as a useful baseline against which future, more refined prompts can be evaluated.

Building structured prompt templates

Once a baseline is established, the next step is to create structured, reusable prompt templates. Templates bring consistency and clarity to the prompting process by ensuring that the model receives all the necessary details to generate high-quality responses.

A well-designed prompt template usually includes the following components:

- **Role:** Defines who or what the model should act as (for example, You are an AI writing assistant).
- **Context:** Provides background or purpose (for example, You are summarizing content for a professional audience).
- **Constraints:** Outlines specific rules or limitations (for example, Keep the response under 50 words and use concise language).
- **Output format:** Specifies how the result should be structured (for example, Provide the response as a short paragraph).

By combining these elements, a simple baseline prompt can be developed into a much more effective and repeatable one. For instance:

"As an AI writing assistant, summarize the key insights from this document for a professional audience. The response must be under 50 words and written in clear, neutral language."

This refined prompt is likely to produce more focused and relevant results. By creating such templates, developers can quickly adapt prompts for different tasks or use cases by changing only the variable details, saving time, and ensuring consistent quality.

Refining prompts through systematic evaluation

Refining prompt templates requires a methodical approach, and one of the most effective techniques for doing this is A/B testing. In this process, two slightly different versions of a prompt—Prompt A and Prompt B—are created and tested to see which one performs better.

For example, a developer might want to know whether explicitly mentioning the target audience improves the quality of the output. They could test two versions:

- Prompt A: "The tone should be neutral and professional."
- Prompt B: "Write in a tone suitable for business readers."

By comparing the results of both prompts against the predefined success criteria, the developer can gather data-driven insights about which phrasing delivers better outcomes. This evidence-based approach helps remove subjectivity from the evaluation process, enabling teams to make measurable improvements over time.

Comparing models in AI Studio

Google AI Studio provides a built-in environment for this kind of evaluation. You can test multiple prompts or even compare how different models perform on the same task. For instance, you might run the same refined prompt using Gemini Pro and Gemini Flash models side by side.

The platform displays the outputs together, along with useful metrics such as token count, response time, and cost. You might observe that Gemini Pro produces slightly more detailed text, while Gemini Flash generates faster, more cost-efficient results.

These comparisons allow teams to make informed, evidence-based choices—balancing creativity, performance, and efficiency—rather than relying on trial and intuition. Over time, this capability supports continuous model selection and prompt optimization across projects.

Comparing model outputs in Google AI Studio



Organizing and reusing tested prompts

As you create and refine multiple prompts, staying organized becomes essential. A prompt library helps maintain this structure. It serves as a centralized collection of high-quality, tested, and reusable prompts—often categorized by task or use case.

A team could maintain different templates for various purposes, such as generating summaries, drafting responses, or writing short informational texts. When a new task arises, you can quickly retrieve a relevant prompt, saving time and ensuring consistent tone and quality across projects. Over time, this library becomes a shared, evolving resource that strengthens collaboration, efficiency, and overall project scalability.

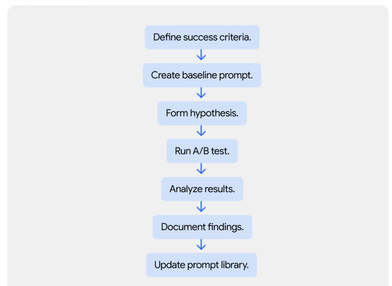
Establishing an evaluation workflow

All the elements discussed—defining success, creating baselines, and testing variations—fit together within a structured evaluation workflow. This workflow provides a repeatable process for prompt optimization:

1. **Define success:** List clear criteria for what a strong output looks like.
2. **Create baseline:** Write the simplest version of the prompt for comparison.
3. **Form a hypothesis:** Identify a weakness in the baseline and propose a change to improve it.
4. **Test the change:** Run an A/B test in AI Studio to compare the original and modified prompts.
5. **Analyze the results:** Evaluate both outputs against the success criteria to see if the change worked.
6. **Document findings:** Record prompts, results, and key insights.
7. **Update the library:** Add successful new prompts to the prompt library as best practices.

Following this structured workflow encourages consistency and objectivity in prompt evaluation. It also allows teams to track progress over time, compare different strategies, and identify which methods yield the highest-quality outcomes.

Prompt evaluation workflow



Documenting and learning from evaluations

Effective documentation is a vital part of the evaluation workflow. Each test—successful or not—provides valuable insights that strengthen future iterations. Carefully record the exact prompt used, the model selected, and any modified parameters such as token limits or context length. You should also save the outputs, note key observations, and write a short analysis explaining what worked, what failed, and why.

Maintaining these detailed records prevents repeated mistakes, promotes transparency, and helps new team members quickly understand project history, decision-making, and rationale behind prompt adjustments. Over time, this documentation evolves into a practical knowledge base that accelerates learning, improves collaboration, and ensures consistent, evidence-backed development across projects. Well-documented evaluation histories also make it easier to standardize quality benchmarks across teams and maintain reproducibility in large-scale AI development environments.

Bringing it all together

The ability to systematically evaluate prompts and models is a core professional skill for AI developers. It transforms prompt creation from guesswork into a disciplined, data-driven engineering process.

Key takeaways

- Effective prompt evaluation relies on structured experimentation and measurable success criteria.
- Baseline prompts and reusable templates create a foundation for systematic improvement.
- A/B testing, documentation, and analysis support evidence-based decision-making.
- Consistent workflows transform AI development into a predictable, repeatable, and scalable practice.

Go to next item ✓ Completed

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